

Donghwan Rho

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Research Interests

- Implementing Language Models Under Homomorphic Encryption
- Mathematical Reasoning in Large Language Models
- Interpretability of Language Models

Education

Ph.D.	Seoul National University , Mathematical Sciences	Mar. 2021 – Present
	• Advisor: Ernest K. Ryu • Expected Graduation: Feb. 2026	
BS	Korea University , Mathematics Education	Mar. 2015 – Feb. 2021
	• Graduated 1st in Class • (Mandatory) Military Service: May 2017 - Jan. 2019 • Teacher's Certificate for Secondary Regular Teacher of Mathematics	

Experience

Research Internship , CryptoLab	Seoul, South Korea
• Homomorphic Encryption Optimization Team	Dec. 2025 -
Workshop Coordinator , SNU-KAIST ML Theory Workshop	Gangneung, South Korea
• Assisted in planning the schedule, selecting the venue, and managing workshop programs.	Aug. 2024
Teaching Practicum	Seoul, South Korea
• Danggok high school	Apr. 2020 - May 2020
Military Service , Auxiliary Police	Seoul, South Korea
• Assisted in maintaining public order and security.	May 2017 – Jan. 2019

Publications and Preprints

*: Equal Contribution

- | | | |
|-----|--|-----------|
| [3] | Studying the Korean Word-Chain Game with RLVR: Mitigating Reward Conflicts via Curriculum Learning
Donghwan Rho
In progress
Paper | Oct. 2025 |
| [2] | Traveling Salesman-Based Token Ordering Improves Stability in Homomorphically Encrypted Language Models
Donghwan Rho* , Sieun Seo*, Hyewon Sung, Chohong Min, Ernest K. Ryu
Submitted
Paper | Oct. 2025 |
| [1] | Encryption-Friendly LLM Architecture
Donghwan Rho* , Taeseong Kim*, Minje Park, Jung Woo Kim, Hyunsik Chae, Ernest K. Ryu, Jung Hee Cheon | Oct. 2024 |

Talks & Presentations

Symposium of the Research Institute of Mathematics, Seoul National University,
Talk

Nov. 2025

- Title: Implementing Efficient Language Models under Homomorphic Encryption
[Site link](#), [Blog post](#) (Thanks to [Yossi Frenkel](#)), [Slides](#)

Korean Mathematical Society Annual Meeting, Talk

Oct. 2025

- Title: Homomorphically encrypted language models stabilized with TSP-sorted tokens
[Link](#)

SNU-Samsung Electronics Collaboration Symposium, Poster Presentation

Aug. 2024

- Title: Encryption-Friendly LLM Architecture

SNU-KAIST ML Theory Workshop, Poster Presentation

Aug. 2024

- Title: Encryption-Friendly LLM Architecture

Honors and Awards

Outstanding TA, Seoul National University [Link](#)

July 2025

Introduction to Computer Programming and Artificial Intelligence for Scientists

University Students Contest of Mathemaitcs, Silver Prize [Link](#)

Dec. 2018

Academic Excellence Scholarship, Korea University

Aug. 2015

Teaching

Peer Tutoring in Basic Sciences, Lead Teaching Assistant

Mar. 2024 - Present

- Represented section TAs to ensure the smooth operation of the peer tutoring program.
- Oversaw tutorings in Mathematics, Physics, Chemistry, Biology, and Statistics.

Basic Calculus, Teaching Assistant

Sep. 2021 - Present

- Matched tutors and tutees for the *Basic Calculus* peer tutoring program.
- Managed program logistics to ensure efficient operation.

SNU Pre-College Program, Head TA, Instructor

- Managed logistics of the mathematics program for freshman undergraduate.
- Gave lectures.
- Winter 2024, Winter 2023, Winter 2022

Introduction to Computer Programming and Artificial Intelligence for Scientists,
Instructor

Spring 2025

- English Course
- Selected as an [outstanding TA](#)

Topics in Machine Intelligence

Fall 2024

- Topics: Continuous-depth neural networks, Diffusion Models, Large Language Models, Vision Language Models, State-space Models
- English Course

- Graduate Course

Mathematical algorithms II

Fall 2022

- Topics: Reinforcement Learning, Diffusion Models, Natural Language Processing
- English Course
- Graduate Course

Calculus 2 Practice, Instructor

- Fall 2021, Fall 2025

Calculus 1 Practice, Instructor

- Spring 2024, Fall 2023, Spring 2023, Spring 2022, Spring 2021

Technical Skills

Languages: Korean (Native), English (Fluent)

Programming Languages: Python, C++

Tools & Technologies: LaTeX, GitHub